

Systems

Storage, execution, consensus, and fault discovery.

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THREE SYSTEMS TO INSPECT

01

raft-mvcc

C++17

598

ASSERTIONS ACROSS SEEDED FAULTS

Five-node failover completes in 11 simulator ticks while minority leaders cannot commit.

FAILURE BOUNDARY

An in-process simulator: no sockets, fsync, snapshots, membership changes, or multi-range transactions.

START `git clone https://github.com/asp53826/raft-mvcc && cd raft-mvcc && make test`

02

columnar-engine

C++17

53.1M

TPC-H Q1 ROWS PER SECOND

6,288 assertions; batched filters are 1.94x the scalar implementation.

FAILURE BOUNDARY

In-memory engine without persistence, distributed execution, or a SQL frontend.

START `git clone https://github.com/asp53826/columnar-engine && cd columnar-engine && make test`

03

dst-harness

Python

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PLANTED DEFECTS CAUGHT

Seeds-to-first-catch range from 1 to 485; the harness also found two real bugs with reproducing seeds.

FAILURE BOUNDARY

The target protocol model is deliberately smaller than a production Raft implementation.

START `git clone https://github.com/asp53826/dst-harness && cd dst-harness && python3 -m venv .venv...`

VERIFICATION CONTRACT

- Claims name the comparison baseline and units.
- Commands reproduce the public correctness path.
- Failure boundaries remain beside the measured result.
- The evidence manifest connects claims to methods and sources.